Chemical Energetics (Summarised)

Energy Changes

Definition: Exothermic change is when thermal energy is given out to surroundings, leading to increase in temperature of surroundings.

- Energy in the form of heat (and/or light) released out to surroundings. Thus, temp of reaction mixture increases, container holding reaction mixture feels warmer to the touch.

Reactants have less energy than the products. There is more energy released to form new bonds compared to energy taken in to break the initial bonds. Thus, there is negative enthalpy change, as there is net (thermal) energy transfer from system to surroundings.

Chemical reactions: - Neutralisation - Combustion - Respiration	Physical processes: - Freezing - Condensation - Deposition of gas to solid (opp. Of sublimation) - Dissolution of acids and alkalis in water - Dissolving sodium carbonate in water

Definition: Endothermic change is when thermal energy is taken in from the surroundings, leading to decrease in temperature of surroundings.

- Energy (often in form of heat and/or light) is absorbed → from surroundings
- Container holding reaction mixture feels cooler to touch

Reactants have less energy than products. There is more energy taken in to break initial bonds compared to energy released to form new bonds. Thus, there is positive enthalpy change, as there is net transfer of (thermal) energy from surroundings to system.

Chemical reactions - Thermal decomposition - Electrolysis of water - Photosynthesis	Physical processes - Boiling - Melting - Evaporation - Sublimation - Dissolution of some ionic compounds in water (e.g. sodium chloride, ammonium chloride, barium nitrate)

Reaction Phases:

In chemical reaction → 2 phases: bond breaking and bong making.

- Bond breaking absorbs energy
- Bond making releases energy (usually in form of thermal energy)

Bond Energy

Definition: Bond energy is the amount of energy absorbed to break one mole of a chemical (covalent) bond. It is also the amount of energy released when one mole of that bond is formed.

- Heat must be absorbed from surroundings to break bonds between 2 atoms in a molecule → bond breaking is endothermic
- Heat is released to surroundings when a covalent bond is formed between 2 atoms → thus bond making is exothermic

Multiple bonds are stronger than single bonds. However, double bonds are not exactly 2 times as strong as single bonds between same 2 atoms.

Activation Energy

- Nonspontaneous reaction is one in which substances do not react spontaneously when mixed.
- This is because → reactants need to have enough energy before they can start reacting.

Definition: Activation Energy is the minimum amount of energy that colliding reactant particles must possess to react with each other.

- Reactant particles with energy greater or equal to the activation energy collide → they are able to break bonds to form new bonds
- Reactant particles with energy less than activation energy collide → they are unable to break bonds → hence no chemical reaction occurs.

<u>**Exo and Endo Thermic Reactions**</u>

Exothermic	Endothermic
Definition: Exothermic reaction is a chemical reaction in which thermal energy is released to the surroundings, leading to an increase in temperature of the surroundings. Reactants have more energy than the products. There is more energy released to form new bonds (to form yyy mol of xxx compound) compared to energy taken in to break initial bonds in yy mol of xxx and xxx. Thus, there is negative enthalpy change, as there is net (thermal) energy transfer from the system to the surroundings.	Definition: chemical reaction in which thermal energy is taken in from the surroundings, leading to decrease in temperature of surroundings. Reactants have less energy than products. There is more energy taken in to break the initial bonds in yyy mol of xxx compound compared to energy released to form new bonds (to form yyy mol of xxx compound). Thus there is positive enthalpy change, as there is net transfer of (thermal) energy from the surroundings to the system.

Explain why maximum temp. is reached when exothermic reaction is competed. - No more thermal energy released as reactants have fully reacted <u>Examples:</u> • Neutralisation reactions • Adding water to anhydrous salts to form hydrated salts. • Displacement of metals • Reaction of reactive metals with water/steam/acids • Corrosion of metals (e.g. rusting) • Combustion of fuels aka burning of fuels in oxygen • Haber process in manufacture of ammonia • respiration	<u>Examples:</u> • thermal decomposition • photosynthesis • formation of nitrogen oxides in a car engine and during thunderstorms when there is lightning • decomposition of CFCs by sunlight

Type of energy change	Exothermic	Endothermic
Direction of heat transfer	Gives out heat from system to surr.	Takes in heat from surr. To system
Energy contents of products compared to reactants	lower	higher
Enthalpy change	Negative	positive
Temp of reaction mixture	increases	decreases

Enthalpy Change

Definition: Enthalpy change refers to the total energy content in a substance. It is represented by symbol H

Enthalpy change, DeltaH measures the difference in energy content of the reactants and products.
Enthalpy change = total energy of products – total energy of reactants
Enthalpy change = total energy taken in to break bonds – total energy released to form new bonds.

Units:
- kilojoules
- kilojoules per mol of reaction KJ/mol

Note: it is not possible to measure energy of substance directly. but it is possible to measure enthalpy change, when substance (reactant) undergoes chemical reaction to form products.

- ⇨ Endothermic: positive enthalpy change → total energy absorbed during bond breaking > total energy released during bond making
- ⇨ Exothermic: negative enthalpy change → total energy absorbed during bond breaking < total energy released during bond making.

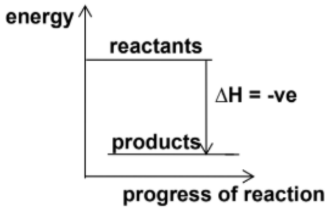
- ➔ Bond breaking phase always has positive enthalpy value and bond making always has negative enthalpy value.

Another formula:
Delta H = delta H of bonding breaking + delta H of bond making
OR
Delta H = total amt of energy absorbed for bond braking + (-total amt of energy released during bond making)

Energy Level Diagrams

Shows overall enthalpy change of reaction

<u>Exothermic Reactions</u>

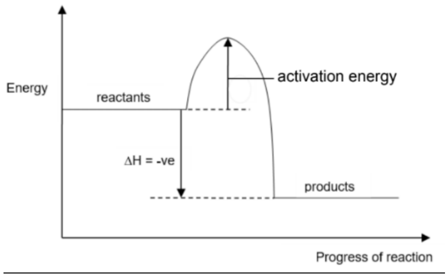


- Gives heat to surroundings → total energy of products will be lower than total energy of reactants.

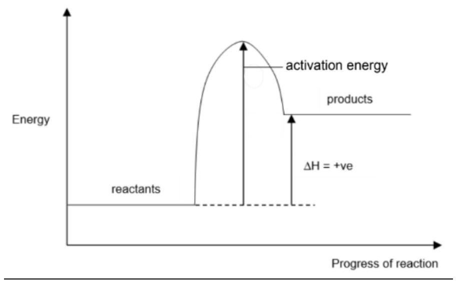
- In exothermic reaction → products are more energetically stable because they have less energy than reactants. Products are more stable at lower energy levels. Thus, many reactions which occurs spontaneously are exothermic reactions

Energy Profile Diagrams

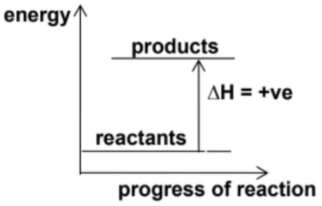
<u>Exothermic Reactions:</u>



<u>Endothermic Reactions</u>



Endothermic Reactions



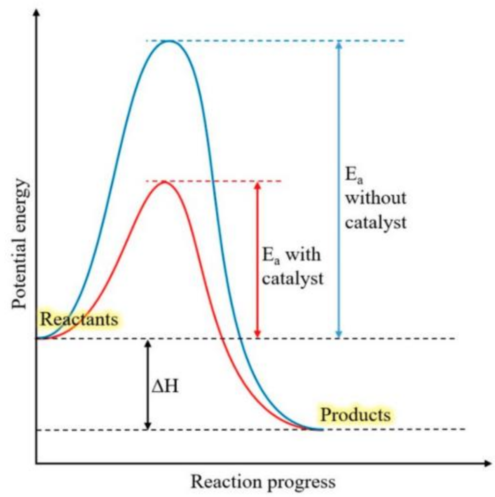
- Endothermic reaction absorbs heat from surroundings → total energy of products will be higher than total energy of the reactants.

- Endothermic reaction → products are less energetically stable → because they have more energy than the reactants. Products are unstable at higher energy levels hence they will break down easily.

Note that the amount of energy released is proportion to amt of substance. if x joules released for 1 mole, then 2x joules released for 2 moles

Catalysed Reaction

- Enthalpy change of uncatalyzed and catalysed reaction remains unchanged, only activation energy changes.



Combustion of Fuels

Fuels → substances that can burn easily in air to give out energy

- Example: fossil fuels such as coal (mainly made up of carbon), petroleum (crude oil) and natural gas (mainly made up of methane) other fuels are petrol and hydrogen gas.

Complete combustion of methane:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

excen

Incomplete combustion of methane:

$$4CH_4(g) + 5O_2(g) \rightarrow 2CO(g) + 8H_2O(l) + 2C(s)$$

Soot

Complete vs incomplete combustion*

Type of Combustion	Complete	Incomplete
Limiting Reactant	Fuel(methane)	Oxygen gas
Excess Reactant	Oxygen gas	Fuel(methane)
Products formed	Carbon dioxide and water	Carbon monoxide, water, Carbon (Soot)
Type of Energy Change	endothermic	exothermic
Amt of Energy released	larger	Lower

During combustion of carbon containing fuels → both CO and CO2 will be formed.

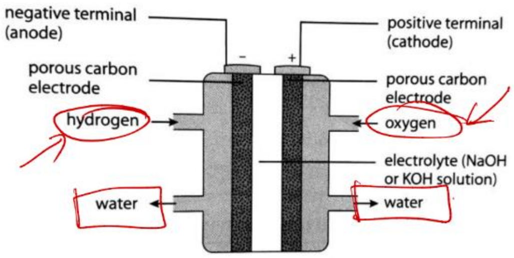
Fuel Cells

Definition: Fuel Cell is a chemical cell in which reactants, usually fuel and oxygen, are continuously supplied to produce electricity directly.

- Hydrogen – oxygen fuel cell → one electrode is supplied with fuel, hydrogen and other electrode is supplied with oxygen. Electrodes are immersed in electrolyte
- Requires constant supply of suitable fuel at anode and an oxidising agent at the cathode
- Hydrogen-oxygen fuel cell → uses hydrogen as fuel and oxygen from air as the oxidising agent. Hydrogen used can be obtained from water by electrolysis, or from hydrocarbons by cracking
- Electrodes used → inert electrodes
- Acidic electrolyte (e.g. acid) or alkaline electrolyte (e.g. potassium hydroxide) can be used.

Chemicals in fuel cell are continuously replaced as they are used up.

This fuel cell is used as a source of electrical power in space vehicles.



The hydrogen–oxygen fuel cell

- Carbon electrodes are porous → allow hydrogen and oxygen gasses → to be in contact with electrolyte → potassium hydroxide solution

	at the negative electrode (anode)	at the positive electrode (cathode)
electrode	At the negative electrode, hydrogen is **oxidised** to form water.	At the positive electrode, oxygen is **reduced** to form hydroxide ions.
reaction	$2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + \mathbf{4e^-}$	$O_2(g) + 2H_2O(l) + \mathbf{4e^-} \rightarrow 4OH^-(aq)$
	Hydrogen reacts with the OH^- ions to form water. The electrons are released at this electrode and move towards the cathode.	Oxygen reacts with water to form OH^- ions. Electrons are received at this electrode from the anode.
overall reaction	$O_2(g) + 2H_2(g) \rightarrow 2H_2O(l)$	
This reaction is equivalent to the combustion of hydrogen. The hydrogen-oxygen fuel cells are used in spacecraft to provide not only electrical energy but also water for the astronauts.		

Advantages	Disadvantages
1. Renewable energy resource 2. Pollution free → only water produced when hydrogen reacts with oxygen 3. Hydrogen → efficient source of energy → provides 2 times	1. Difficult to find cheap source of hydrogen. Obtaining hydrogen from electrolysis is an expensive process. 2. Hydrogen is extremely flammable + explosive. Special precautions

amount of energy compared to same quantity of many other fuels.	must be taken in the storage and transport of the gas 3. Hydrogen exists as a gas → and thus requires larger storage volume compared to other fuels which are usually liquids.

Explaining whether reaction is exo or endo

1. State that bond breaking is endothermic and bond making is exothermic
2. State energy of products is higher/lower than reactants
3. State whether energy taken in from surr to system for breaking bonds in xx mol of xxx compound is more/less than energy released from system to surr to make new bonds to form xxx mol of xxx compound
4. Net energy transfer from system to surr / surr to system
5. Positive/negative enthalpy change
6. Thus endo / exo

Imp to state direction of movement → taken in or given out